Engineering Chemistry · Unit_05 · Engineering Chemistry - Lab Component

DRAFT — UE26CY151A · Unit_05 — Engineering Chemistry - Lab Component · Complete Unit Master

Source-complete searchable master · transcript, slide text, page visuals and archive references

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Search this single unit master by lesson, timestamp, slide text or source page. Read the transcript in timestamp order, then compare extracted slide text with its page-faithful source figure. Formula candidates and archive question mappings are explicitly flagged for review. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · UE26CY151A · Unit_05: Engineering Chemistry - Lab Component
2. 2 · Historical question-source pages · unit mapping candidates
3. Appendix A · Symbol glossary
4. Appendix B · Sources, coverage and gaps
5. Appendix C · Revision and retrieval practice

M1

Unit_05 · Engineering Chemistry - Lab Component

1 UE26CY151A · Unit_05: Engineering Chemistry - Lab Component

This unit master organizes source material for Engineering Chemistry, Unit_05, officially titled Engineering Chemistry - Lab Component. The official catalogue includes this unit, but the current course bucket has no matching unit folder or unit-specific lecture files. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Official catalogue entry: UE26CY151A, Unit_05 (Engineering Chemistry - Lab Component). Bucket folder present: no. Videos: 0; slide decks: 0. No unit-specific current lecture sources are present in the bucket.

EV0000001

2 Historical question-source pages · unit mapping candidates

This section preserves historical paper/question-bank pages whose extracted terms matched Engineering Chemistry, Unit_05 (Engineering Chemistry - Lab Component) above the dataset's candidate threshold. The association has not been manually confirmed, source text/OCR may be incomplete, and the pages are not rewritten or presented as solved questions. EV0000002 · EV0000003

Important points and why

1. Archive page 1 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. Department of Science and Humanities ENGINEERING CHEMISTRY EV0000002

2. Archive page 2 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY • INTRODUCTION • COMPUTATIONAL METHODS • APPLICATIONS EV0000003
3. Archive page 3 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY Introduction Computational chemistry is a set of techniques for the investigating chemical problems on a computer rather than using chemicals. It uses the results of theoretical chemistry incorporated into efficient computer programs to calculate the properties of a molecule. The first theoretical investigation was done by Walter Heitler and Fritz London in 1927 lecture_notes.dvi EV0000004
4. Archive page 4 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY 1998 Walter Kohn : For the development of density functional theory. John A. Pople : For the development of computational methods in quantum chemistry. EV0000005
5. Archive page 5 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY 2013 For the development of computer based methods to model complex systems. EV0000006
6. Archive page 6 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY Objective of computational chemistry: To solve chemistry problems by simulating chemical systems (molecular, biological, materials) to provide reliable, accurate and comprehensive information at an atomic level. Advantages of computational chemistry: ☒Calculations are easy to perform than carrying experiments. ☒Calculations are less costly, whereas experiments are becoming more expensive ☒Calculations are safe than performing experiments. lecture_notes.dvi EV0000007
7. Archive page 9 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY Drug discovery: Computational chemistry plays a pivotal role in design and screening of potential drug candidates by simulating the interactions between the drug molecules and biological targets. Paracetamol All these spectra are generated using Gaussion 16 EV0000008
8. Archive page 11 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. Department of Science and Humanities ENGINEERING CHEMISTRY EV0000009
9. Archive page 13 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY • Molecular mechanics: ☒Molecular mechanics programs use equations based on classical physics to calculate force fields. ☒It is based on the model of a molecule as a collection of balls(atoms) held together by springs(bonds). ☒Its too fast like a very large molecules like steroids can be optimized in seconds and allows geometry optimization . EV0000010

10. Archive page 14 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY • Molecular dynamics: ☒Molecular dynamics is a molecular mechanics program designed to mimic the movement of atoms within a molecule. The laws of motion to molecule is the base for it. ☒Molecular dynamics can be carried out on a molecule to generate different conformation which on energy minimization. Staggered vs Eclipsed Conformations of Ethane - Master Organic Chem EV0000011
11. Archive page 15 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY • Semi empirical methods: ☒Semi empirical calculations are based on Schrödinger equation. ☒It represents a middle road between the mostly qualitative results available from molecular mechanics and the high computationally demanding quantitative results from Ab initio methods. ☒Capable of calculating transition states and excited states. EV0000012
12. Archive page 16 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY Ab initio calculations: ☒Ab initio calculations are based on Schrödinger equation and it describes, among other things, how the electrons in a molecule behave. ☒The wave function is a mathematical function that can be used to calculate the electrons distribution. EV0000013
13. Archive page 17 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY • Density functional theory ☒DFT are like Ab initio and semi empirical calculations, based on Schrödinger equation. ☒However, unlike the other two methods, DFT does not calculate a conventional wave function , but rather derives the electron distribution directly. ☒Density functional calculations are usually faster than Ab initio, but slower than semi empirical method. EV0000014
14. Archive page 19 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY ☒Computational chemistry plays a pivotal role in advancing our understanding of chemical systems and accelerating scientific discovery. ☒Its advantages in cost-effectiveness, efficiency, safety, accessibility, versatility, and accuracy make it an indispensable tool in modern research and development. ☒Computational chemistry is not a replacement for experimental studies, but plays an important role in enabling chemists to design and develop the desired chemicals/ products. Conclusion. Ref. Book: Introduction to computational chemistry by Frank Jensen, Wiley publications EV0000015
15. Archive page 23 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. Hydrogen Energy • The Hydrogen energy has been identified as one of potential sources of fuel in the state-of-art technological era • Hydrogen production, storage and transportation has become the focus of science and technology • Sustainable production of Hydrogen from renewable feedstocks and electrolysis of water has taken lead role in technology development • More emphasis is provided to understanding and engineering the production and storage of Hydrogen produced by Electrolysis method 3 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage EV0000016

16. Archive page 25 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. Hydrogen Economy • Hydrogen used as fuel reduces pollution • It is carbon free energy source • Hydrogen economy is a vision of an energy delivery infrastructure based on hydrogen as a carbon-free energy carrier • It provides • alternative energy economy in the form of a parallel power and transportation infrastructure • technical solutions for energy-efficient hydrogen production, storage technology and delivery infrastructure 5 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage EV0000017
17. Archive page 28 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. • Photolysis • Processes use light energy to split water into hydrogen and oxygen • One of the potential for sustainable hydrogen production • Low environmental impact • Yet in early stage of research • Electrolysis • uses electricity to split water into hydrogen and oxygen • Electrolysis reaction takes place in a unit called an electrolyzer • Method is promising option for carbon-free hydrogen production from renewable and nuclear resources. 8 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage EV0000018
18. Archive page 30 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. • Shift reforming (Water Gas shift reactions) CO + H2O → CO2 + H2 • Catalyst: Fe3O4/Cr2O3 • Exothermic reaction (ΔHo = -41.0 KJ/mole) • Favoured at low temperature (350oC) • H2 produced will have traces of Carbon monoxide as impurities • Purification • CO2 is removed to produce pure Hydrogen • Traces of CO is removed by methanation (350 – 450oC) • CO + 3H2O ⊠CH4 + H2O 10 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage EV0000019
19. Archive page 31 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. Green Hydrogen refers to production of Hydrogen gas from renewable resources Green Hydrogen production will help us reach both the momentum and promise for greener mobility and industry Hydrogen production through Electrolysis produces clean energy with pollution free process by using electricity Hydrogen production by electrolysis can be achieved by three methods. 1. Alkaline Electrolysis 2. Proton Membrane Electrolysis 3. High temperature Electrolysis Green Hydrogen production 11 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage EV0000020
20. Archive page 32 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. • In Alkaline electrolyte process • aqueous solution containing either NaOH or KOH used • typical concentration of 20-40 wt % • operation temperatures are between 343 – 363 K • operating pressure up to 3 MPa • Charge carrier: OH- • Anodic materials: Ni and its alloys • Cathodic materials: Cd, Pb, Cu, Ag, Pt, Pd. • Electrolyte materials : KOH, NaOH • Catalyst materials: Ni-Zn, Ni-Co-Au, RuO2, LaCoO3 12 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage EV0000021
21. Archive page 34 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. • Zero emission of CO2 (if this process combined with renewable energy sources like electricity form solar or wind energy) • Alkaline water electrolysis is conventional methods with easiest, simplest, and suitable methods for Hydrogen production • Electrolysis requires needs Direct current supply • Electrolyser split water into Hydrogen and Oxygen gas 14 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage EV0000022

22. Archive page 35 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. • Splitting of water is very small approximately 7- 10 moles/ litre • because pure water is the very poor conductor of electricity hence acid (H2SO4 in acid electrolyzer) or base (KOH or NaOH in Alkaline electrolyzer) is used to improve the conductivity • Solutions split into ions which improving conductivity of the electrolyte. • The Reaction of water splitting is as follows: • At Anode: 2OH- ☒½ O2 + H2O + 2 e- • At Cathode: 2H2O + 2e- ☒H2 + 2OH- • The basic equation of water splitting is represented as: 2H2O (liquid) + Energy ☒2H2(gas) + O2 (gas) 15 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage EV0000023

23. Archive page 36 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. Conclusion: • The Hydrogen energy has been identified as one of potential sources of fuel, Production of Green Hydrogen is the key focus to meet the sustainable development. • Hydrogen is not available as a pure gas, it must be produced from other compounds like fossil fuels, biomass, electrolysis, etc., • Electrolysis of water to produce hydrogen is one of the most viable technology • Hydrogen evolution takes place at cathode materials, whereas oxygen evolution takes place at anode materials. • Catalysts improve the performance of the electrode materials to generate high and pure hydrogen gas. 16 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage EV0000024

24. Archive page 39 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. Module content: • Hydrogen Storage • Methods of Storage 2 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage EV0000025

25. Archive page 40 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. Hydrogen Energy • The Hydrogen storage is crucial for effective utilization of Hydrogen for its versatile applications. • Methods used for hydrogen storage: • Liquid form • Gaseous • Solid State 3 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage EV0000026

26. Archive page 41 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. • Hydrogen Storage in Metal-Organic Frameworks: A Review - research journal (gyanvihar.org) 4 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage EV0000027

27. Archive page 42 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. 5 • Hydrogen can be physically stored as either a gas or a liquid • Storage of hydrogen in the form of gas usually requires high-pressure tanks (350–700 bar tank pressure) • Storage of hydrogen as a liquid requires cryogenic temperatures because the boiling point of hydrogen is −252.8°C at an atmospheric pressure • Hydrogen can also be stored on the surfaces of solids (adsorption) or within solids (absorption) 5 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage very low EV0000028

28. Archive page 43 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. Solid state hydrogen storage: • Depending on the type of hydrogen being stored in host materials, solid-state hydrogen storage can be classified into two categories: • Chemisorption of hydrogen atoms • Physisorption of intact hydrogen molecules • In chemisorption (absorption into matter), binding of hydrogen atoms occur into the surface • In case of physisorption (adsorption on surface).The Hydrogen molecule is bound onto the surface of adsorbent 6 ENGINEERING CHEMISTRY Hydrogen Energy Production and Storage EV0000029

29. Archive page 1 from 5__1pIeTzocnU.pdf; unit match candidate score 1.0; source text/OCR may be partial. 11/11/24, 11:34 AM ESA -JAN 2024 - UE23CY151A (set- 2) about:blank 1/7 (6.0 Marks) (6.0 Marks) (6.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - JAN 2023 UE23CY151A - Engineering Chemistry Total Marks : 100.0 1.a. For a rotating diatomic molecule, i) Derive an expression for moment of inertia 'I' for the molecule. ii) If B =10 cm , give the rotational energy levels of this molecule, in cm for rotational quantum numbers J=0,1,2,3, and draw an energy level diagram for the same molecule up to J = 3. 1.b. 1.c. (i) Draw a neat labeled phase diagram of the Carbon dioxide system specifying the different areas and lines. Give temperature and pressure values at the triple point and critical point. (ii) What is the Reduced phase rule? Why is general Gibb's Phase rule not applied to the 2-component system? -1 -1 EV0000030

SHORT-NOTE RECAP

29 historical archive pages are included as question-source references. Inspect the original paper before relying on OCR or unit placement; no answers are newly authored here.

EV0000002

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Extracted historical pages: Chem_Extra_Topics__1-HfOeZNwo.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Extra/Chem_Extra_Topics__1-HfOeZNwo.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 5__1pIeTzocnU.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/5__1pIeTzocnU.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. Unit_05 is listed in the official catalogue but has no matching source folder in PESU_Materials.

2. No current video or slide materials were available for this catalogue-only unit; archive excerpts appear only when the unit-match candidate is strong.
3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

Coverage

Every inventoried source unit has an explicit coverage decision. Covered means reviewed and mapped to content; exclusions and gaps remain visible.

Evidence	Source unit	Status	Decision
EV0000001	SRC00001/catalogue	covered	Unit title and index copied from the official units.json catalogue; human content review is pending.
EV0000002	SRC00002/q1	covered	Original archive PDF page 1; candidate unit association is text-based and unconfirmed.
EV0000003	SRC00002/q2	covered	Original archive PDF page 2; candidate unit association is text-based and unconfirmed.
EV0000004	SRC00002/q3	covered	Original archive PDF page 3; candidate unit association is text-based and unconfirmed.
EV0000005	SRC00002/q4	covered	Original archive PDF page 4; candidate unit association is text-based and unconfirmed.
EV0000006	SRC00002/q5	covered	Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
EV0000007	SRC00002/q6	covered	Original archive PDF page 6; candidate unit association is text-based and unconfirmed.
EV0000008	SRC00002/q9	covered	Original archive PDF page 9; candidate unit association is text-based and unconfirmed.
EV0000009	SRC00002/q11	covered	Original archive PDF page 11; candidate unit association is text-based and unconfirmed.
EV0000010	SRC00002/q13	covered	Original archive PDF page 13; candidate unit association is text-based and unconfirmed.
EV0000011	SRC00002/q14	covered	Original archive PDF page 14; candidate unit association is text-based and unconfirmed.

Evidence	Source unit	Status	Decision
EV0000012	SRC00002/q15	covered	Original archive PDF page 15; candidate unit association is text-based and unconfirmed.
EV0000013	SRC00002/q16	covered	Original archive PDF page 16; candidate unit association is text-based and unconfirmed.
EV0000014	SRC00002/q17	covered	Original archive PDF page 17; candidate unit association is text-based and unconfirmed.
EV0000015	SRC00002/q19	covered	Original archive PDF page 19; candidate unit association is text-based and unconfirmed.
EV0000016	SRC00002/q23	covered	Original archive PDF page 23; candidate unit association is text-based and unconfirmed.
EV0000017	SRC00002/q25	covered	Original archive PDF page 25; candidate unit association is text-based and unconfirmed.
EV0000018	SRC00002/q28	covered	Original archive PDF page 28; candidate unit association is text-based and unconfirmed.
EV0000019	SRC00002/q30	covered	Original archive PDF page 30; candidate unit association is text-based and unconfirmed.
EV0000020	SRC00002/q31	covered	Original archive PDF page 31; candidate unit association is text-based and unconfirmed.
EV0000021	SRC00002/q32	covered	Original archive PDF page 32; candidate unit association is text-based and unconfirmed.
EV0000022	SRC00002/q34	covered	Original archive PDF page 34; candidate unit association is text-based and unconfirmed.
EV0000023	SRC00002/q35	covered	Original archive PDF page 35; candidate unit association is text-based and unconfirmed.
EV0000024	SRC00002/q36	covered	Original archive PDF page 36; candidate unit association is text-based and unconfirmed.
EV0000025	SRC00002/q39	covered	Original archive PDF page 39; candidate unit association is text-based and unconfirmed.
EV0000026	SRC00002/q40	covered	Original archive PDF page 40; candidate unit association is text-based and unconfirmed.
EV0000027	SRC00002/q41	covered	Original archive PDF page 41; candidate unit association is text-based and unconfirmed.

Evidence	Source unit	Status	Decision
EV00000028	SRC00002/q42	covered	Original archive PDF page 42; candidate unit association is text-based and unconfirmed.
EV00000029	SRC00002/q43	covered	Original archive PDF page 43; candidate unit association is text-based and unconfirmed.
EV00000030	SRC00003/q1	covered	Original archive PDF page 1; candidate unit association is text-based and unconfirmed.

C Revision and retrieval practice